

Module 3 COMSOL FEM Implementation Note

Click-by-Click Beginner Tutorial for RadiationEffects_proj2

Parallel COMSOL implementation of two-dimensional defect diffusion, reaction, annealing, and verification

Purpose of this note. This document explains, at beginner click-by-click level, how to implement the Module 3 finite-element defect-evolution model in COMSOL Multiphysics. The main project solver remains the MATLAB FEM implementation because it exposes the mesh, element mass matrix, diffusion matrix, reaction matrix, assembly logic, and time-stepping system directly. The COMSOL model is a parallel commercial-FEM reference model. It is useful for validating the governing equation, checking mesh convergence, inspecting diffusion and annealing behavior, and building intuition before coupling Module 3 to Modules 2, 4, 5, and 6. The instructions are written in the style of a COMSOL tutorial: what to click, what to type, why each step is being done, what result to expect, and what error usually means.

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Symbols and acronyms used in this document

Symbol	Meaning	Units
C	Defect concentration field solved in Module 3. In the baseline COMSOL model this is the scalar dependent variable.	m^{-3}
C_i	Concentration of defect species i , such as vacancy, self-interstitial, divacancy, vacancy-oxygen complex, or an effective charged trap family.	m^{-3}
C_V	Vacancy concentration, used in optional multi-species examples.	m^{-3}
C_I	Self-interstitial concentration, used in optional multi-species examples.	m^{-3}
$\mathbf{J}_C$	Defect flux.	$\text{m}^{-2} \text{s}^{-1}$
D	Defect diffusivity.	m^2/s
D_0	Arrhenius diffusion prefactor.	m^2/s
E_D	Defect migration activation energy.	J or eV
k_{ann}	First-order annealing, trapping, or removal coefficient.	s^{-1}
k_0	Annealing prefactor or attempt frequency.	s^{-1}
E_{ann}	Annealing activation energy.	J or eV
S	Volumetric defect source term.	$\text{m}^{-3} \text{s}^{-1}$
T	Temperature field. In this tutorial it is first constant, then optionally imported from Module 4.	K
Ω	Two-dimensional silicon computational domain.	m^2
$\partial\Omega$	Boundary of the two-dimensional computational domain.	m
$\hat{\mathbf{n}}$	Outward unit normal vector on the domain boundary.	1
L_x, L_y	Width and height of the 2D silicon domain.	m
h	Representative mesh element size.	m
N_a	Linear finite-element shape function associated with local node a .	1
M	FEM mass matrix for transient storage.	m^2
K_D	FEM diffusion matrix.	m^2/s
K_R	FEM reaction or annealing matrix.	m^2/s
Δt	Time-step size or output-time spacing.	s

1 What Module 3 does in the project

Module 3 evolves radiation-induced defect populations after energy deposition and defect creation. In the project logic, Module 1 provides an energy-deposition or damage-generation map, Module 3 turns that initial damage into a time-dependent defect concentration field, Module 2 uses the charged portion of those defects to update electrostatics, Module 4 can provide temperature-dependent diffusivity and annealing rates, and Module 5 uses the resulting fields and defect populations for carrier transport and recombination.

The key physical question for Module 3 is:

Given an initial spatial distribution of radiation-induced defects in silicon, how does that distribution spread, anneal, react, or transform with time?

The baseline one-species equation is

$$\frac{\partial C}{\partial t} = \nabla \cdot (D \nabla C) - k_{\text{ann}} C + S. \quad (1.1)$$

Here D controls defect migration, k_{ann} controls local removal or transformation, and S is an optional volumetric source. If the radiation event is modeled as an initial condition, then after $t = 0$ one usually sets $S = 0$.

The corresponding defect flux is

$$\mathbf{J}_C = -D \nabla C. \quad (1.2)$$

A default closed-domain boundary condition is zero normal flux,

$$\hat{\mathbf{n}} \cdot \mathbf{J}_C = 0, \quad \text{or equivalently} \quad \hat{\mathbf{n}} \cdot D \nabla C = 0. \quad (1.3)$$

This means that defects diffuse within the simulated silicon region but do not leave through the modeled boundaries.

Important distinction. Module 3 does not solve electrostatic potential or electric field. It solves the material-defect concentration field. If some defect species are charged, Module 2 later maps $C_i(x, y, t)$ into defect charge density $\rho_{\text{def}} = q \sum_i z_i C_i$. Thus Module 3 supplies the evolving defect populations; Module 2 translates those populations into electrostatic source terms.

2 Why COMSOL is useful even though MATLAB remains the main tool

The MATLAB FEM implementation is the primary project path because it exposes the numerical method explicitly. In MATLAB, the project can show exactly how the triangular mesh is created, how each element matrix is computed, how element matrices are assembled, how boundary conditions are imposed, and how the time-dependent matrix system is solved.

COMSOL is useful for a different reason. It provides a mature finite-element environment where the same PDE can be entered through a standard interface, meshed, solved, visualized, and verified rapidly. The goal is not to replace the MATLAB implementation. The goal is to build a parallel reference model.

The relationship is:

$$\boxed{\text{MATLAB} = \text{main transparent solver}}, \quad \boxed{\text{COMSOL} = \text{parallel FEM reference and visualization tool}}. \quad (2.1)$$

For interview and project-defense purposes, this is a strong position. It shows that the physics does not depend on one software package. The same continuum model can be implemented explicitly in MATLAB and independently in a commercial FEM tool.

3 Which COMSOL physics interface to use

There are two practical COMSOL routes.

3.1 Recommended route: Coefficient Form PDE

The recommended route is **Coefficient Form PDE**. It is usually found under:

Mathematics → **PDE Interfaces** → **Coefficient Form PDE**.

This route is the best match to the FEM derivation because it lets the user enter the coefficients of the PDE directly. It also avoids requiring the Chemical Species Transport Module.

The general COMSOL coefficient-form equation is commonly written as

$$e_a \frac{\partial^2 u}{\partial t^2} + d_a \frac{\partial u}{\partial t} + \nabla \cdot (-c \nabla u - \alpha u + \gamma) + \beta \cdot \nabla u + a u = f. \quad (3.1)$$

For Module 3, the dependent variable is $u = C$. Rewrite Eq. (1.1) as

$$\frac{\partial C}{\partial t} + \nabla \cdot (-D \nabla C) + k_{\text{ann}} C = S. \quad (3.2)$$

Therefore the coefficient mapping is

$$\boxed{e_a = 0, \quad d_a = 1, \quad c = D, \quad a = k_{\text{ann}}, \quad f = S,} \quad (3.3)$$

with

$$\boxed{\alpha = 0, \quad \beta = 0, \quad \gamma = 0.} \quad (3.4)$$

3.2 Alternative route: Transport of Diluted Species

If the COMSOL installation includes the Chemical Species Transport interfaces, one can also use **Transport of Diluted Species**. That interface is more physically named for concentration transport, but the coefficient-form PDE route is more general and closer to the FEM equations used in this project. This document therefore treats Coefficient Form PDE as the official project tutorial path and gives the diluted-species route only as an optional alternative.

Sign warning. In Coefficient Form PDE, the annealing term goes on the left as $+aC$. Therefore use $a = k_{\text{ann}}$, not $a = -k_{\text{ann}}$. If the pure-annealing test grows instead of decays, the reaction sign is wrong.

4 Finite-element form behind the COMSOL model

COMSOL hides the element matrices, but it is solving the same weak-form FEM system that is written explicitly in the MATLAB notes. Multiply Eq. (1.1) by a test function w and integrate over the domain:

$$\int_{\Omega} w \frac{\partial C}{\partial t} d\Omega = \int_{\Omega} w \nabla \cdot (D \nabla C) d\Omega - \int_{\Omega} w k_{\text{ann}} C d\Omega + \int_{\Omega} w S d\Omega. \quad (4.1)$$

Integrate the diffusion term by parts:

$$\int_{\Omega} w \frac{\partial C}{\partial t} d\Omega + \int_{\Omega} D \nabla w \cdot \nabla C d\Omega + \int_{\Omega} w k_{\text{ann}} C d\Omega = \int_{\Omega} w S d\Omega + \int_{\partial\Omega} w D \nabla C \cdot \hat{\mathbf{n}} d\Gamma. \quad (4.2)$$

For zero-flux boundaries, $D \nabla C \cdot \hat{\mathbf{n}} = 0$, so the boundary term vanishes. The weak form is

$$\boxed{\int_{\Omega} w \frac{\partial C}{\partial t} d\Omega + \int_{\Omega} D \nabla w \cdot \nabla C d\Omega + \int_{\Omega} w k_{\text{ann}} C d\Omega = \int_{\Omega} w S d\Omega.} \quad (4.3)$$

For a linear triangular element,

$$C^{(e)}(x, y, t) = \sum_{a=1}^3 N_a(x, y) C_a^{(e)}(t). \quad (4.4)$$

The element arrays are

$$M_{ab}^{(e)} = \int_{\Omega_e} N_a N_b d\Omega, \quad (4.5)$$

$$K_{D,ab}^{(e)} = \int_{\Omega_e} D \nabla N_a \cdot \nabla N_b d\Omega, \quad (4.6)$$

$$K_{R,ab}^{(e)} = \int_{\Omega_e} k_{\text{ann}} N_a N_b d\Omega, \quad (4.7)$$

$$f_a^{(e)} = \int_{\Omega_e} N_a S d\Omega. \quad (4.8)$$

After global assembly,

$$M \frac{d\mathbf{C}}{dt} + (K_D + K_R)\mathbf{C} = \mathbf{f}. \quad (4.9)$$

A backward-Euler time step gives

$$\boxed{[M + \Delta t(K_D + K_R)] \mathbf{C}^{n+1} = M\mathbf{C}^n + \Delta t\mathbf{f}^{n+1}.} \quad (4.10)$$

This is the matrix system that COMSOL assembles internally.

5 Tutorial model definition

The tutorial builds a 2D silicon rectangle with an initial Gaussian defect cloud. The cloud diffuses and anneals in time. This simple geometry is deliberate. It makes the physical behavior easy to check before using imported radiation maps or coupled fields.

The demonstration values below are chosen to make the model visibly evolve over a reasonable simulation window. They are not asserted as calibrated defect constants for a specific silicon device.

Parameter	Type exactly this	Meaning
Lx	10 [um]	Domain width
Ly	5 [um]	Domain height
T0	300 [K]	Baseline temperature
kB	1.380649e-23 [J/K]	Boltzmann constant
eV	1.602176634e-19 [J]	Electron-volt conversion
D0	1e-8 [m^2/s]	Diffusion prefactor
ED	0.45	Migration energy in eV as a dimensionless number
k0	1e6 [1/s]	Annealing prefactor

Parameter	Type exactly this	Meaning
Eann	0.65	Annealing energy in eV as a dimensionless number
Cbase	0[1/m ³]	Uniform background defect concentration
Cpeak	1e21[1/m ³]	Peak initial defect concentration
x0	0.5*Lx	Initial cloud center in x
y0	0.5*Ly	Initial cloud center in y
sigx	0.8[um]	Initial cloud width in x
sigy	0.5[um]	Initial cloud width in y
dtout	1e-6[s]	Output time spacing
tend	1e-4[s]	Final simulation time

The component variables are

$$T_{\text{field}} = T_0, \quad (5.1)$$

$$D_{\text{def}} = D_0 \exp(-(E_D \cdot \text{eV}) / (k_B \cdot T_{\text{field}})), \quad (5.2)$$

$$k_{\text{ann}} = k_0 \exp(-(E_{\text{ann}} \cdot \text{eV}) / (k_B \cdot T_{\text{field}})), \quad (5.3)$$

$$C_{\text{init}} = C_{\text{base}} + C_{\text{peak}} \exp(-((x-x_0)^2 / (2 \cdot \text{sigx}^2) + (y-y_0)^2 / (2 \cdot \text{sigy}^2))), \quad (5.4)$$

$$S_{\text{def}} = 0[1/(\text{m}^3 \cdot \text{s})]. \quad (5.5)$$

6 Beginner click-by-click implementation

6.1 Start COMSOL and open the Model Wizard

1. Open COMSOL Multiphysics from the desktop, Start menu, or application launcher.
2. When the startup screen appears, click **Model Wizard**.
3. If COMSOL instead opens a blank model, choose **File** → **New**; then click **Model Wizard**.

6.2 Choose the model dimension

1. In **Select Space Dimension**, click **2D**.
2. Click **Next**.

The project model is two-dimensional because it represents lateral variation and depth variation in a silicon cross section. A 1D model can verify simple formulas, but it cannot represent laterally localized radiation damage.

6.3 Add the Coefficient Form PDE physics

1. In **Select Physics**, expand **Mathematics**.
2. Expand **PDE Interfaces**.
3. Click **Coefficient Form PDE**.
4. Click **Add**.
5. Look at the lower part of the Model Wizard where the selected physics appears.
6. If COMSOL displays the default dependent variable name, change it to **C**.
7. If COMSOL shows a field-name box, set the field name to **C** as well.
8. Click **Next**.

Why the dependent variable is named C. Module 3 solves a concentration field. Naming the dependent variable **C** makes later expressions readable: $d(C, x)$ means $\partial C / \partial x$, $d(C, y)$ means $\partial C / \partial y$, and **C** itself is the defect concentration.

6.4 Choose a time-dependent study

1. In **Select Study**, click **Time Dependent**.
2. Click **Done**.

Module 3 is a transient problem because defect populations evolve with time. A stationary study would only be appropriate for a final equilibrium or steady-state limit, not for the radiation damage evolution model.

6.5 Save the model immediately

1. Choose **File** → **Save As**.
2. Save the model as `module3_defect_diffusion_reaction_baseline.mph`.
3. Use a working folder outside Git if the file is large. For documentation purposes, the project stores only the PDF and TeX instructions.

Saving early prevents loss of setup work and gives COMSOL a known path for later exports.

7 Set global units and parameters

7.1 Check the unit system

1. In the **Model Builder**, click the top model node. This may be named **Untitled.mph**, **Model 1**, or the saved filename.
2. In the **Settings** window, find **Unit System**.
3. Select **SI**.

Use SI units throughout the model: meters, seconds, kelvin, joules, and number concentration in m^{-3} .

7.2 Open the Parameters table

1. In **Model Builder**, right-click **Global Definitions**.
2. Click **Parameters**.
3. A node named **Parameters 1** should appear under **Global Definitions**.
4. Click **Parameters 1**.
5. In the table, enter the rows below.

Name	Expression	Description
Lx	10 [um]	Silicon domain width
Ly	5 [um]	Silicon domain height
T0	300 [K]	Constant baseline temperature
kB	1.380649e-23 [J/K]	Boltzmann constant
eV	1.602176634e-19 [J]	Electron-volt in joules
D0	1e-8 [m^2/s]	Defect diffusion prefactor
ED	0.45	Migration barrier measured in eV
k0	1e6 [1/s]	Annealing prefactor
Eann	0.65	Annealing barrier measured in eV
Cbase	0 [1/m^3]	Background defect concentration

Name	Expression	Description
Cpeak	1e21 [1/m ³]	Initial Gaussian peak concentration
x0	0.5*Lx	Damage center x-location
y0	0.5*Ly	Damage center y-location
sigx	0.8 [um]	Gaussian width in x
sigy	0.5 [um]	Gaussian width in y
dtout	1e-6 [s]	Output time spacing
tend	1e-4 [s]	Final time

7.3 Confirm parameter units

1. After typing the parameters, check the **Value** and **Unit** columns that COMSOL computes.
2. Confirm that Lx and Ly convert to meters.
3. Confirm that D0 has units m²/s.
4. Confirm that k0 has units 1/s.
5. Confirm that Cpeak has units 1/m³.

If COMSOL marks an expression orange or red, click the warning icon and read the unit message. Most errors at this stage are missing brackets, such as typing 10um instead of 10[um].

8 Build the 2D silicon geometry

8.1 Create a rectangle

1. In **Model Builder**, expand **Component 1**.
2. Right-click **Geometry 1**.
3. Click **Rectangle**.
4. In the **Settings** window for Rectangle, find **Size and Shape**.
5. Set **Width**: Lx.
6. Set **Height**: Ly.

7. Find **Position**.
8. Set **x**: 0.
9. Set **y**: 0.
10. Click **Build Selected**.
11. If the rectangle does not appear, click **Build All**.

The rectangle is the silicon computational domain. The lower left corner is at $(0, 0)$, and the upper right corner is at (L_x, L_y) .

8.2 Check the geometry scale

1. In the graphics window, click **Zoom Extents** if needed.
2. Confirm that the rectangle is much wider than it is tall, with aspect ratio $L_x/L_y = 2$.
3. If the geometry appears empty, go back to Geometry 1 and check that the units were entered with brackets.

9 Define variables for coefficients, initial condition, and diagnostics

9.1 Create domain variables

1. In **Model Builder**, right-click **Component 1**.
2. Click **Definitions** if it is not already visible.
3. Right-click **Definitions**.
4. Choose **Variables**.
5. Click the new **Variables 1** node.
6. In the **Settings** window, set **Geometric Entity Level** to **Domain**.
7. Select the rectangular silicon domain in the graphics window or select domain number 1 in the selection list.

9.2 Type the variable definitions

Enter the following rows in the Variables table.

Name	Expression	Description
Tfield	T0	Temperature used by Module 3
Ddef	$D0 \cdot \exp(-(ED \cdot \text{eV}) / (kB \cdot Tfield))$	Defect diffusivity
kann	$k0 \cdot \exp(-(Eann \cdot \text{eV}) / (kB \cdot Tfield))$	Annealing coefficient
Cinit	$Cbase + Cpeak \cdot \exp(-((x-x0)^2 / (2 \cdot sigx^2) + (y-y0)^2 / (2 \cdot sigy^2)))$	Initial defect cloud
Sdef	$0 [1 / (m^3 \cdot s)]$	Continuing source term
Jcx	$-Ddef \cdot d(C, x)$	x-component of defect flux
Jcy	$-Ddef \cdot d(C, y)$	y-component of defect flux
Jcmag	$\text{sqrt}(Jcx^2 + Jcy^2)$	Defect flux magnitude
Crel	$C / Cpeak$	Normalized concentration

Why use variables instead of typing everything directly into the PDE interface?

Variables make the model readable and less error-prone. If the diffusivity model changes later, only Ddef needs to be edited. If Module 4 supplies a temperature field later, only Tfield needs to be changed.

9.3 Check the Arrhenius expressions

The diffusivity expression is

$$D = D_0 \exp\left(-\frac{E_D}{k_B T}\right). \quad (9.1)$$

In COMSOL, the numerator is typed as ED*eV because ED is entered as a numerical value in electron-volts. The expression

$$(ED \cdot \text{eV}) / (kB \cdot Tfield) \quad (9.2)$$

is dimensionless. The same logic applies to $E_{ann} \cdot eV / (kB \cdot T_{field})$.

10 Enter the Coefficient Form PDE

10.1 Open the coefficient settings

1. In **Model Builder**, expand **Component 1**.
2. Expand **Coefficient Form PDE**.
3. Click the domain equation node, often called **Coefficient Form PDE 1**.
4. In the **Settings** window, find the coefficient fields.
5. Make sure the rectangular domain is selected.

10.2 Enter the PDE coefficients

Set the coefficients exactly as below.

COMSOL coefficient	Type this	Physical meaning
ea	0	No second-order time derivative
da	1	Coefficient of $\partial C / \partial t$
c	Ddef	Diffusion coefficient
alpha	0	No conservative drift in baseline
gamma	0	No imposed conservative flux
beta	0	No nonconservative advection in baseline
a	kann	First-order annealing loss
f	Sdef	Volumetric source

If COMSOL expects the diffusion coefficient c as a 2D tensor, enter

$$\text{Ddef}, 0; 0, \text{Ddef}. \quad (10.1)$$

If it accepts a scalar coefficient, **Ddef** is sufficient and represents isotropic diffusion.

10.3 Check that the PDE matches the target equation

With these entries, COMSOL solves

$$\frac{\partial C}{\partial t} + \nabla \cdot (-D_{\text{def}} \nabla C) + k_{\text{ann}} C = S_{\text{def}}. \quad (10.2)$$

Moving the diffusion and reaction terms to the right gives

$$\frac{\partial C}{\partial t} = \nabla \cdot (D_{\text{def}} \nabla C) - k_{\text{ann}} C + S_{\text{def}}, \quad (10.3)$$

which is exactly the Module 3 baseline equation.

11 Set initial and boundary conditions

11.1 Set the initial defect concentration

1. Under **Coefficient Form PDE**, click **Initial Values**.
2. In the **Settings** window, find the field for the initial value of **C**.
3. Type **Cinit**.

The simulation now starts from a localized radiation-damage cloud. At early time, the maximum concentration is near (x_0, y_0) . As time evolves, the cloud spreads and decays.

11.2 Use zero-flux boundaries

For Coefficient Form PDE, homogeneous natural flux is usually the default. To make the model explicit:

1. Expand **Coefficient Form PDE**.
2. Look for a boundary node named **Zero Flux**, **Flux/Source**, or **Default Boundary Condition**. The exact label depends on the COMSOL version.
3. Click that boundary-condition node.
4. Select all four outer boundaries of the rectangle.
5. If the node is **Zero Flux**, leave it unchanged.
6. If the node is **Flux/Source**, set the flux/source entry to 0.

The intended condition is

$$\hat{\mathbf{n}} \cdot (-D\nabla C) = 0. \quad (11.1)$$

This prevents defects from crossing the boundary. It is the correct default for the pure diffusion inventory-conservation test.

11.3 Optional absorbing surface boundary

Do not use this in the default tutorial. It is an optional variant.

1. Right-click **Coefficient Form PDE**.
2. Choose **Dirichlet Boundary Condition**.
3. Select the boundary that acts as a perfect sink, for example the top boundary.
4. Set **C** to $0[1/m^3]$.

This represents an ideal surface that removes defects. It will intentionally destroy total inventory conservation.

12 Mesh the domain

12.1 Create a triangular mesh

1. In **Model Builder**, right-click **Mesh 1**.
2. Choose **Free Triangular**.
3. Right-click **Mesh 1** again.
4. Choose **Size**.
5. Click the **Size** node.
6. In **Element Size**, choose **Custom**.
7. Enter **Maximum element size**: $\min(L_x, L_y)/40$.
8. Enter **Minimum element size**: $\min(L_x, L_y)/200$.
9. Enter **Maximum element growth rate**: 1.3.
10. Enter **Curvature factor**: 0.3.
11. Enter **Resolution of narrow regions**: 1.
12. Click **Build All**.

12.2 Why mesh resolution matters

The mesh must resolve the initial Gaussian cloud. A rough criterion is that several elements should fit across σ_x and σ_y . If the cloud width is $0.5 \mu\text{m}$ and the largest element is comparable to that width, the initial condition is under-resolved. Under-resolved initial damage maps produce artificial numerical diffusion.

12.3 Perform a first visual mesh check

1. After clicking **Build All**, inspect the mesh in the graphics window.
2. Confirm that the triangles are reasonably uniform.
3. Confirm that the domain contains enough elements near the Gaussian cloud center.
4. If the mesh is too coarse, reduce the maximum element size to $\min(L_x, L_y)/80$ and rebuild.

13 Set the time-dependent study and solver

13.1 Set output times

1. In **Model Builder**, expand **Study 1**.
2. Click **Step 1: Time Dependent**.
3. In the **Settings** window, find **Study Settings**.
4. In the **Times** box, type `range(0,dtout,tend)`.

This stores the solution at times $0, \Delta t_{\text{out}}, 2\Delta t_{\text{out}}, \dots, t_{\text{end}}$.

13.2 Inspect time-solver settings

1. Expand **Study 1**.
2. Expand **Solver Configurations**.
3. Expand **Solution 1**.
4. Click **Time-Dependent Solver**.
5. If a **Time Stepping** method is visible, choose **BDF**.
6. Set **Relative tolerance** to $1\text{e-}3$ for the first run.

7. For verification runs, later tighten the relative tolerance to $1\text{e-}5$.

Diffusion-reaction equations can be stiff, especially when annealing is fast or the mesh is fine. An implicit BDF time integrator is appropriate for this class of problem.

13.3 Compute the solution

1. Click the **Study 1** node.
2. Click **Compute**.
3. Wait for the solver to finish.
4. If COMSOL reports a unit warning, read it before ignoring it.
5. If the solver fails, check the coefficient signs, units in the exponentials, and mesh size.

14 Plot and interpret the results

14.1 Surface plot of defect concentration

1. Right-click **Results**.
2. Choose **2D Plot Group**.
3. Rename the plot group **Defect concentration**.
4. Right-click **Defect concentration**.
5. Choose **Surface**.
6. Click the **Surface** node.
7. In **Expression**, type **C**.
8. In **Unit**, use $1/\text{m}^3$ if needed.
9. Click **Plot**.
10. Use the time selector to compare $t = 0$ and later times.

Expected behavior: the Gaussian concentration peak decreases, the profile broadens, and the total inventory decays only if annealing is active.

14.2 Normalized concentration plot

1. Right-click **Results**.
2. Choose **2D Plot Group**.
3. Rename it **Normalized concentration**.
4. Add a **Surface** plot.
5. Set **Expression**: C_{rel} .
6. Click **Plot**.

This plot is useful when the absolute concentration values are very large. It shows the spatial shape independent of the initial scale.

14.3 Defect flux arrow plot

1. Right-click **Results**.
2. Choose **2D Plot Group**.
3. Rename it **Defect flux arrows**.
4. Right-click the plot group.
5. Choose **Arrow Surface**.
6. For the x-component, type J_{cx} .
7. For the y-component, type J_{cy} .
8. Click **Plot**.

The arrows should point from high concentration toward low concentration. For a centered Gaussian cloud, the arrows should point outward early in time.

14.4 Flux magnitude plot

1. Create another **2D Plot Group**.
2. Rename it **Defect flux magnitude**.
3. Add a **Surface** plot.
4. Set **Expression**: J_{mag} .

5. Click **Plot**.

The flux magnitude is largest on the sides of the concentration peak, not exactly at the peak. At the peak, $\nabla C \approx 0$, so the diffusive flux is small there.

14.5 Line plot through the damage cloud

1. Right-click **Results**.
2. Choose **1D Plot Group**.
3. Rename it **Centerline concentration**.
4. Right-click the plot group.
5. Choose **Line Graph**.
6. In the **Data** section, choose a cut line dataset if available. If not, create one using the next steps.
7. To create a cut line, right-click **Results** → **Datasets**.
8. Choose **Cut Line 2D**.
9. Set **Point 1**: 0, y0.
10. Set **Point 2**: Lx, y0.
11. Return to the **Line Graph** and select this cut line dataset.
12. Set **Expression**: C.
13. Click **Plot**.

This plot makes the diffusion broadening easy to see quantitatively.

15 Derived values and quantitative diagnostics

15.1 Total defect inventory

1. Right-click **Results**.
2. Choose **Derived Values** → **Surface Integration**.
3. Select the rectangular domain.

4. Set **Expression**: C .
5. Make sure the dataset is the time-dependent solution.
6. Click **Evaluate**.

The quantity is

$$M_C(t) = \int_{\Omega} C(x, y, t) d\Omega. \quad (15.1)$$

Because this is a 2D cross-sectional model, the value has units of defects per meter out-of-plane thickness if C is volumetric concentration.

15.2 Peak concentration

1. Right-click **Results**.
2. Choose **Derived Values** → **Maximum** → **Surface Maximum** if available.
3. Select the domain.
4. Set **Expression**: C .
5. Click **Evaluate**.

The peak concentration should decrease with time because diffusion spreads the cloud and annealing removes defects.

15.3 Mean-square spread

For a more advanced diagnostic, define approximate spatial moments. Create derived values using surface integration of:

$$C \quad \text{for inventory,} \quad (15.2)$$

$$x \cdot C \quad \text{for first moment in } x, \quad (15.3)$$

$$y \cdot C \quad \text{for first moment in } y, \quad (15.4)$$

$$(x-x_0)^2 \cdot C \quad \text{for spread in } x, \quad (15.5)$$

$$(y-y_0)^2 \cdot C \quad \text{for spread in } y. \quad (15.6)$$

Pure diffusion should increase the spread while preserving inventory under zero-flux boundaries.

16 Export data for MATLAB comparison

16.1 Export solution data on the mesh

1. Right-click **Results**.
2. Choose **Export** → **Data**.
3. Click the new **Data** export node.
4. Select the time-dependent dataset.
5. In **Expressions**, add C , J_{cx} , J_{cy} , and J_{cmag} .
6. In **Filename**, type `module3_comsol_defect_solution.csv`.
7. Click **Export**.

16.2 Compare with MATLAB

A fair comparison should use either common derived quantities or interpolation onto a shared grid. Recommended quantities are:

- total defect inventory $\int C \, d\Omega$,
- peak concentration $\max(C)$,
- centerline profile $C(x, y_0, t)$,
- mean-square spread,
- late-time exponential decay rate when annealing dominates.

Do not compare raw nodal values unless the MATLAB and COMSOL meshes are identical or one solution has been interpolated consistently onto the other mesh.

17 Verification tutorials

17.1 Verification 1: uniform-state preservation

Purpose: verify that the solver does not create artificial spatial structure.

1. Go to **Global Definitions** → **Parameters 1**.

2. Set **Cpeak** to $0 [1/m^3]$.
3. Set **Cbase** to $1e20 [1/m^3]$.
4. Set **k0** to $0 [1/s]$.
5. Click **Compute**.
6. Plot **C**.
7. Use **Surface Maximum** and **Surface Minimum** for **C**.

Expected result:

$$C(x, y, t) = C_0. \quad (17.1)$$

The maximum and minimum should be nearly identical.

17.2 Verification 2: pure diffusion inventory conservation

Purpose: verify the diffusion operator and zero-flux boundary condition.

1. Set **Cpeak** back to $1e21 [1/m^3]$.
2. Set **Cbase** to $0 [1/m^3]$.
3. Set **k0** to $0 [1/s]$.
4. Keep all boundaries as zero flux.
5. Click **Compute**.
6. Evaluate **intop1(C)** if you created an integration operator, or use **Surface Integration** of **C** at all times.

Expected result:

$$\frac{d}{dt} \int_{\Omega} C d\Omega = 0. \quad (17.2)$$

The cloud spreads, but total inventory stays constant to numerical tolerance.

17.3 Verification 3: pure annealing

Purpose: verify the sign and magnitude of the reaction term.

1. Set **D0** to $0 [m^2/s]$.
2. Set **Cpeak** to $0 [1/m^3]$.

3. Set **Cbase** to $1\text{e}20[1/\text{m}^3]$.
4. Set **k0** to $1\text{e}6[1/\text{s}]$.
5. Click **Compute**.
6. Plot or evaluate the domain average of **C** versus time.

The exact solution is

$$C(t) = C(0) \exp(-k_{\text{ann}} t). \quad (17.3)$$

If concentration increases, the sign of **a** is wrong.

17.4 Verification 4: diffusion plus annealing

Purpose: verify that diffusion spreads the profile while annealing removes inventory.

1. Restore **D0** to $1\text{e}-8[\text{m}^2/\text{s}]$.
2. Use the Gaussian initial condition.
3. Use nonzero **k0**.
4. Compute.
5. Plot **C** at early, middle, and final times.
6. Evaluate total inventory versus time.

Expected result: the concentration cloud broadens and total inventory decreases. If k_{ann} is spatially uniform and boundaries are zero flux, total inventory should decay approximately as

$$M_C(t) = M_C(0) \exp(-k_{\text{ann}} t). \quad (17.4)$$

17.5 Verification 5: mesh refinement

Purpose: confirm that the result is not a mesh artifact.

1. Run the baseline model with maximum element size $\min(\text{Lx}, \text{Ly})/40$.
2. Record total inventory, peak concentration, and centerline profile at final time.
3. Change maximum element size to $\min(\text{Lx}, \text{Ly})/80$.
4. Rebuild the mesh and recompute.

5. Repeat with $\min(L_x, L_y)/120$ if needed.
6. Compare the metrics.

A trustworthy model should change less as the mesh is refined. For interview explanation, say that mesh refinement checks spatial discretization error.

18 Adding integration operators for easier diagnostics

1. In **Model Builder**, right-click **Definitions** under **Component 1**.
2. Choose **Component Couplings** → **Integration**.
3. Rename the operator `intopC`.
4. Select the rectangular domain.
5. In **Results**, use expressions such as `intopC(C)`, `intopC(x*C)`, and `intopC((x-x0)^2*C)`.

This makes it easier to track conservation and diffusion spread without repeatedly creating surface integration nodes.

19 Optional extension: spatially varying temperature from Module 4

19.1 Analytic temperature field test

Before importing actual Module 4 data, test spatially varying temperature analytically.

1. Go to **Definitions** → **Variables 1**.
2. Replace `Tfield=T0` with:

$$T0 + 50[\text{K}] * \exp(-((x-x0)^2/(2*\text{sigx}^2) + (y-y0)^2/(2*\text{sigy}^2))). \quad (19.1)$$

3. Keep `Ddef` and `kann` unchanged.
4. Recompute.

Because D and k_{ann} are Arrhenius functions, warmer regions can have faster diffusion and stronger annealing. This mimics one-way coupling from Module 4.

19.2 Import a temperature map

1. Right-click **Global Definitions**.
2. Choose **Functions** → **Interpolation**.
3. Rename the function **Tmod4**.
4. In **Settings**, choose the data source: **File**.
5. Browse to a CSV file containing columns such as **x**, **y**, and **T**.
6. Set the number of arguments to 2 if the file is spatial only.
7. Set argument units to **m** and **m**.
8. Set function unit to **K**.
9. Click **Import** or **Plot** to check the function.
10. In **Variables 1**, set **Tfield=Tmod4(x,y)**.
11. Recompute.

This implements a one-way Module 4 → Module 3 coupling.

20 Optional extension: imported radiation damage map

The Gaussian initial condition is a verification-friendly stand-in for a radiation damage map. For project realism, Module 1 or a damage conversion step can provide an imported initial condition.

1. Prepare a CSV file with columns **x**, **y**, and **Cinit_map**.
2. Right-click **Global Definitions**.
3. Choose **Functions** → **Interpolation**.
4. Rename the function **Cmap**.
5. Set the file as the data source.
6. Set argument units to **m**, **m**.
7. Set function unit to **1/m³**.
8. In **Variables 1**, set **Cinit=Cmap(x,y)**.
9. Recompute.

When using imported maps, always plot the initial condition before time stepping. If the imported map is shifted, transposed, or unit-scaled incorrectly, the simulation may run but represent the wrong physics.

21 Optional extension: charged-defect drift from Module 2

The baseline Module 3 model is diffusion plus reaction. If charged-defect drift under an electric field is included, the flux can be written as

$$\mathbf{J}_C = -D\nabla C + \mu_d z_d q C \mathbf{E}. \quad (21.1)$$

Using the Einstein relation for a dilute charged species,

$$\mu_d = \frac{D}{k_B T}, \quad (21.2)$$

where the force is $z_d q \mathbf{E}$.

In COMSOL Coefficient Form PDE, the conservative flux is

$$-c\nabla u - \alpha u + \gamma. \quad (21.3)$$

To represent drift velocity $\mathbf{v}_d = \mu_d z_d q \mathbf{E}$, use

$$\mathbf{J} = -D\nabla C + \mathbf{v}_d C. \quad (21.4)$$

Since COMSOL's flux form is $-D\nabla C - \alpha C$, set

$$\alpha = -\mathbf{v}_d. \quad (21.5)$$

This extension should be added only after the baseline diffusion-reaction model passes all verification tests.

22 Optional multi-species implementation

A more complete Module 3 can evolve multiple defect species:

$$\frac{\partial C_V}{\partial t} = \nabla \cdot (D_V \nabla C_V) + G_V - R_V, \quad (22.1)$$

$$\frac{\partial C_I}{\partial t} = \nabla \cdot (D_I \nabla C_I) + G_I - R_I. \quad (22.2)$$

For example, vacancy-interstitial recombination can use

$$R_{VI} = k_{VI}C_VC_I. \quad (22.3)$$

Then

$$\frac{\partial C_V}{\partial t} = \nabla \cdot (D_V \nabla C_V) - R_{VI}, \quad (22.4)$$

$$\frac{\partial C_I}{\partial t} = \nabla \cdot (D_I \nabla C_I) - R_{VI}. \quad (22.5)$$

In COMSOL:

1. In the Model Wizard, when adding Coefficient Form PDE, set the number of dependent variables to 2 if the interface asks.
2. Name the variables **CV** and **CI**.
3. Define variables **DV**, **DI**, and **RVI**.
4. For the **CV** equation, set **c=DV**, **a=0**, and **f=-RVI** if using the left-side coefficient convention carefully.
5. For the **CI** equation, set **c=DI**, **a=0**, and **f=-RVI**.
6. Check signs by verifying that recombination decreases both species.

Because multi-species coefficient entry can vary by COMSOL version, the recommended project path is to validate the scalar equation first, then add a second equation only after the one-species model is fully understood.

23 Alternative implementation using Transport of Diluted Species

If available, the **Transport of Diluted Species** interface can implement the same scalar equation.

1. Choose **Model Wizard**.
2. Select **2D**.
3. Expand **Chemical Species Transport**.
4. Select **Transport of Diluted Species**.
5. Choose **Time Dependent**.
6. Build the same rectangle.

7. Define the same parameters and variables.
8. Set the diffusion coefficient to `Ddef`.
9. Add a reaction term with expression `-kann*C + Sdef`, adapting `C` to the interface's default concentration variable name if needed.
10. Set initial concentration to `Cinit`.
11. Use no-flux boundaries on all sides.
12. Mesh, solve, plot, and verify as above.

This interface is more intuitive for species transport, but Coefficient Form PDE is preferred for transparent mapping to the FEM weak form.

24 Beginner workflow checks before moving to advanced variants

24.1 Expected Model Builder tree

After the baseline model is built, the Model Builder should contain the following essential structure:

```

Global Definitions
  Parameters 1
Component 1
  Definitions
    Variables 1
  Geometry 1
    Rectangle 1
  Coefficient Form PDE
    Coefficient Form PDE 1
    Initial Values
    Zero Flux or Flux/Source boundary node
  Mesh 1
    Free Triangular 1
    Size 1
Study 1
  Step 1: Time Dependent
Results
  Defect concentration
  Defect flux arrows
  Defect flux magnitude

```


If your model tree looks very different, the model can still be valid, but you should understand why. For a beginner workflow, this tree is a useful checklist because it ties every physics decision to a visible COMSOL node.

24.2 Save a clean baseline model before modifying it

1. Choose **File** → **Save As**.
2. Save the verified baseline as `module3_comsol_baseline_verified.mph`.
3. Before adding temperature coupling, imported maps, drift, or multiple species, choose **File** → **Save As** again.
4. Use a new filename such as `module3_comsol_temperature_variant.mph`.

This prevents a common beginner problem: a working baseline model is overwritten by an experimental coupled model, and later it is unclear which change introduced an error.

24.3 Create a simple animation of concentration spreading

1. Make sure the **Defect concentration** plot group exists and plots **C**.
2. In the top ribbon or Results toolbar, look for **Animation**.
3. If not visible, right-click **Results** and look for an **Animation** option.
4. Choose a player or file animation depending on the COMSOL version.
5. Set the plot group to **Defect concentration**.
6. Set the frames to use the stored time-dependent solution times.
7. Play the animation.
8. Check that the defect cloud spreads smoothly and does not jump, oscillate, or develop non-physical negative regions.

The animation is not only for presentation. It is a diagnostic. A physically reasonable diffusion-reaction model should evolve smoothly. Sudden jumps often indicate a time-stepping, parameter, or source-term problem.

24.4 Export a plot image for documentation

1. Right-click **Results**.
2. Choose **Export** → **Image**.
3. Select the plot group, for example **Defect concentration**.
4. Choose a filename such as `module3_comsol_C_final.png`.
5. Set the resolution high enough for documentation, for example 300 dpi if the option is available.
6. Click **Export**.

This step is optional for the project package because the present note intentionally avoids embedding images. Still, exporting images is useful when comparing COMSOL results against MATLAB plots during development.

25 Parameter sweep tutorial for sensitivity checks

A simple parameter sweep helps determine whether the model responds physically to changes in diffusion and annealing parameters.

25.1 Sweep the annealing prefactor

1. In **Study 1**, right-click the study node.
2. Choose **Parametric Sweep**.
3. Click the new **Parametric Sweep** node.
4. In **Parameter name**, choose or type `k0`.
5. In **Parameter value list**, type `1e4[1/s] 1e5[1/s] 1e6[1/s]`.
6. Keep the time-dependent step active.
7. Click **Compute**.
8. Plot total inventory or peak concentration for each parameter value.

Expected behavior: larger `k0` increases `kann`, causing faster defect removal and faster decay of total inventory.

25.2 Sweep the diffusion prefactor

1. In the **Parametric Sweep** settings, replace the parameter with D_0 .
2. Use values such as $1e-10[m^2/s]$ $1e-9[m^2/s]$ $1e-8[m^2/s]$.
3. Compute again.
4. Compare the final concentration profiles.

Expected behavior: larger D_0 produces faster spreading. If the peak decays but the spatial width does not change, then annealing is dominating over diffusion or the diffusion coefficient is too small for the chosen time window.

26 Creating a short COMSOL report for your own records

Some COMSOL installations include a report-generation feature. If available:

1. In the top menu, choose **File** → **Report** or right-click **Reports** if the Reports branch is visible.
2. Choose a brief model report rather than a full exhaustive report.
3. Include parameters, geometry, physics settings, mesh statistics, solver settings, and selected plots.
4. Export the report as PDF or HTML for your own notes.

The project repository should still keep the hand-written LaTeX note as the authoritative explanation, because it includes the physics reasoning and exact project interpretation. The COMSOL report is a useful audit trail of the software settings.

27 Common mistakes and diagnostic checks

Symptom	Likely cause and correction
Concentration grows during pure annealing	Reaction sign is wrong. Use $a=k_{ann}$ in Coefficient Form PDE.
Total inventory changes during pure diffusion with zero-flux boundaries	Boundary is not zero flux, or integration is not over the full domain. Check boundary nodes and use surface integration of C .

Symptom	Likely cause and correction
Initial Gaussian looks blocky	Mesh is too coarse compared with sigx and sigy . Refine the mesh.
Solution changes strongly under mesh refinement	Spatial discretization error is still significant. Refine until inventory, peak value, and centerline profile converge.
Unit warning in exponential	Confirm that $ED \cdot eV / (kB \cdot T_{field})$ and $E_{ann} \cdot eV / (kB \cdot T_{field})$ are dimensionless.
Negative concentration appears	Time tolerance may be loose, mesh may be coarse, or source/reaction signs may be wrong. Tighten tolerance and check signs.
Flux arrows point toward the peak	Flux sign is wrong. Use $J_{cx} = -D_{def} \cdot d(C, x)$ and $J_{cy} = -D_{def} \cdot d(C, y)$.
No visible diffusion	The Arrhenius coefficient may be extremely small. Evaluate Ddef numerically and adjust demonstration parameters if needed.
Everything decays to zero too fast	kann is too large for the chosen time window. Check kann*tend .
COMSOL cannot parse $d(C, x)$	The dependent variable may not be named C . Check the PDE dependent-variable name.

28 Final checklist before trusting the model

Before using the COMSOL result as a reference for MATLAB, check all items below.

1. The model dimension is 2D.
2. The physics interface is Coefficient Form PDE.
3. The dependent variable is **C**.
4. The PDE coefficients are **da=1**, **c=Ddef**, **a=kann**, and **f=Sdef**.
5. The reaction sign passes the pure-annealing test.
6. The boundary condition is zero flux for conservation tests.
7. The mesh resolves the initial damage cloud.
8. The time-dependent study uses **range(0,dtout,tend)**.

9. Pure diffusion conserves total inventory.
10. Pure annealing matches exponential decay.
11. The flux arrows point down the concentration gradient.
12. Exported quantities are compared to MATLAB using integral metrics or interpolation onto a common grid.

29 How to explain this COMSOL model in an interview

A concise explanation is:

Module 3 solves a time-dependent defect diffusion-reaction equation. In COMSOL I implement it through the Coefficient Form PDE interface by writing the equation as $\partial C / \partial t + \nabla \cdot (-D \nabla C) + k_{\text{ann}} C = S$. The coefficient entries are $d_a = 1$, $c = D$, $a = k_{\text{ann}}$, and $f = S$. The zero-flux boundary condition represents a closed silicon region and is essential for checking inventory conservation in the pure diffusion limit. COMSOL then performs the same finite-element operations as the MATLAB code: it meshes the 2D domain, interpolates C with shape functions, assembles the mass, diffusion, and reaction matrices, and advances the transient system with an implicit solver. I verify the model using uniform-state preservation, pure diffusion conservation, pure annealing exponential decay, and mesh refinement.

30 Expanded beginner tutorial layer: exact COMSOL reproduction workflow

This section raises the Module 3 COMSOL note to the same tutorial density as the Module 4 heat-transfer note. The goal is not merely to state the governing equation. The goal is to let a beginner reproduce the model by following the text line by line.

How to use this section. Build only the one-species defect diffusion and annealing model first. Do not begin with multiple defect species, temperature maps, radiation maps, or electric-field drift. Those extensions are meaningful only after the baseline transient diffusion-reaction model passes the inventory, annealing, and diffusion verification tests.

30.1 First model variant to build

The first COMSOL model should solve

$$\frac{\partial C}{\partial t} = \nabla \cdot (D \nabla C) - k_{\text{ann}} C + S, \quad (30.1)$$

with constant D , constant k_{ann} , optional source S , and zero-flux boundaries. The recommended first file name is `module3_defect_diffusion_baseline.mph`.

In this baseline version, C may be interpreted as one effective defect concentration, for example a vacancy concentration C_V , but the mathematics is kept single-species.

30.2 Detailed Model Wizard sequence

1. Open COMSOL Multiphysics.
2. Click **Model Wizard**.
3. Under **Select Space Dimension**, choose **2D**.
4. Click **Next**.
5. In the physics search box, type **Coefficient Form PDE**.
6. Select **Mathematics > PDE Interfaces > Coefficient Form PDE (c)**.
7. Click **Add**.
8. In the dependent-variable field, set the dependent variable name to **C**. If COMSOL initially uses **u**, replace it with **C**.
9. Click **Study**.
10. Choose **Time Dependent**.
11. Click **Done**.
12. Save the model as `module3_defect_diffusion_baseline.mph`.

Why Coefficient Form PDE? Module 3 is a generic diffusion-reaction equation for a defect concentration, not heat transfer and not electrostatics. The Coefficient Form PDE interface lets us type the mass coefficient, diffusion coefficient, reaction coefficient, source term, initial condition, and flux boundary condition directly. This is closest to the MATLAB FEM form.

30.3 Expected Model Builder tree

After the Model Wizard, check that the Model Builder contains:

- **Global Definitions**
- **Component 1 (comp1)**
 - **Definitions**
 - **Geometry 1**
 - **Coefficient Form PDE (c)**
 - **Mesh 1**
- **Study 1**
- **Results**

If the dependent variable is not **C**, change it before building the rest of the model. Consistent variable naming avoids confusion when exporting results to MATLAB.

30.4 Global parameter table: exact entries

Right-click **Global Definitions** and choose **Parameters**. Enter the following table.

Name	Expression	Description
Lx	10[um]	Domain width.
Ly	5[um]	Domain height.
Ddef	1e-14[m ² /s]	Baseline defect diffusivity.
kann	1e3[1/s]	Baseline first-order annealing rate.
Cbg	0[1/m ³]	Background defect concentration.
Cpeak	1e22[1/m ³]	Peak initial defect concentration.
x0c	0.5*Lx	Initial hotspot center in x .
y0c	0.5*Ly	Initial hotspot center in y .
sigx	0.7[um]	Initial Gaussian width in x .
sigy	0.7[um]	Initial Gaussian width in y .
S0	0[1/(m ³ *s)]	Volumetric defect source after $t = 0$.

Name	Expression	Description
tend	5e-6[s]	Final simulation time.
dtout	1e-7[s]	Output time spacing.

30.5 Geometry: exact rectangle steps

1. Click **Component 1 > Geometry 1**.
2. Set **Length unit** to **um**.
3. Right-click **Geometry 1**.
4. Choose **Rectangle**.
5. For **Width**, type **Lx**.
6. For **Height**, type **Ly**.
7. For position, use **x=0** and **y=0**.
8. Click **Build Selected**.
9. Click **Zoom Extents**.

The rectangle is the reduced 2D silicon domain. Zero-flux boundary conditions on this rectangle mean that defects remain inside the modeled region unless an absorbing or reactive boundary is intentionally added later.

30.6 Definitions and variables: exact entries

Right-click **Component 1 > Definitions** and choose **Variables**. Rename the node **var_defect_model**. Type the following rows.

Name	Expression	Description
Cinit	$C_{bg} + C_{peak} \exp\left(-\left(\frac{(x-x_0)^2}{2 \cdot \sigma_x^2} + \frac{(y-y_0)^2}{2 \cdot \sigma_y^2}\right)\right)$	Initial defect concentration cloud.
Duse	Ddef	Diffusivity used in PDE.
kuse	kann	Annealing rate used in PDE.

Name	Expression	Description
Suse	S0	Source term used in PDE.
Jx_def	-Duse*Cx	Defect flux in x if COMSOL exposes Cx.
Jy_def	-Duse*Cy	Defect flux in y if COMSOL exposes Cy.

If Cx and Cy are not recognized in your COMSOL version, use d(C,x) and d(C,y) instead. The flux definitions then become -Duse*d(C,x) and -Duse*d(C,y).

30.7 Coefficient Form PDE: exact coefficient entries

Click **Coefficient Form PDE (c)**. Then click its main domain equation node, often named **Coefficient Form PDE 1**. Make sure **Domain 1** is selected.

COMSOL's coefficient form may appear as

$$e_a \frac{\partial^2 u}{\partial t^2} + d_a \frac{\partial u}{\partial t} + \nabla \cdot (-c \nabla u - \alpha u + \gamma) + \beta \cdot \nabla u + a u = f. \quad (30.2)$$

For the Module 3 baseline, set the coefficients as follows.

Coefficient	Type this	Meaning
ea	0	No second time derivative.
da	1	Storage term $\partial C / \partial t$.
c	Duse	Diffusive coefficient.
alpha	0	No conservative advection in baseline.
gamma	0	No imposed flux source in baseline.
beta	0	No nonconservative advection in baseline.
a	kuse	First-order removal term.
f	Suse	Volumetric generation source.

With these entries, COMSOL solves

$$\frac{\partial C}{\partial t} + \nabla \cdot (-D \nabla C) + kC = S, \quad (30.3)$$

which is equivalent to

$$\frac{\partial C}{\partial t} = \nabla \cdot (D \nabla C) - kC + S. \quad (30.4)$$

Sign check. The annealing term should reduce C . Therefore the coefficient **a** should be positive for decay. If you put **-kuse** in **a**, the solution will grow exponentially rather than anneal.

30.8 Initial values: exact clicks

1. Under **Coefficient Form PDE (c)**, click **Initial Values 1**.
2. In the field for **C**, type **Cinit**.
3. Confirm that **Domain 1** is selected.

The initial defect distribution is now a Gaussian cloud superimposed on optional background concentration.

30.9 Boundary conditions: exact zero-flux setup

Coefficient Form PDE usually gives a default zero-flux boundary if no boundary source is added. To verify:

1. Expand **Coefficient Form PDE (c)**.
2. Look for the default boundary node. It may be named **Flux/Source 1** or similar.
3. Select all four boundaries.
4. Set the boundary flux/source term to 0.

The target boundary condition is

$$\hat{\mathbf{n}} \cdot (-D \nabla C) = 0, \quad (30.5)$$

meaning no defect inventory leaves the computational domain through the boundary.

30.10 Mesh: exact steps and physical meaning

1. Click **Mesh 1**.
2. Change **Sequence type** to **User-controlled mesh** if available.
3. Right-click **Mesh 1**.

4. Choose **Free Triangular**.
5. Right-click **Mesh 1** and choose **Size**.
6. Choose **Custom**.
7. Set **Maximum element size** to $\min(\text{sigx}, \text{sigy})/4$.
8. Set **Minimum element size** to $\min(\text{sigx}, \text{sigy})/20$.
9. Click **Build All**.

For diffusion problems, the mesh must resolve the initial spatial gradients. If the Gaussian damage cloud is much narrower than the element size, COMSOL will smear or underresolve the initial concentration.

30.11 Time-dependent study: exact entries

1. Click **Study 1 > Step 1: Time Dependent**.
2. In **Times**, type `range(0,dtout,tend)`.
3. Expand **Study Settings** if needed.
4. Confirm that **Coefficient Form PDE (c)** is selected.
5. Click **Study 1**.
6. Click **Compute**.
7. Save after a successful solve.

If the solver fails, reduce `dtout`, reduce `Ddef`, refine the mesh, or inspect whether the coefficient signs have accidentally made the solution unstable.

31 Expanded postprocessing and diagnostics for Module 3

31.1 Surface plot of concentration

1. Right-click **Results**.
2. Choose **2D Plot Group**.
3. Rename it `Defect_concentration_C`.
4. Right-click the plot group and choose **Surface**.

5. In **Expression**, type **C**.
6. In the plot group settings, choose a later time from the **Time** list.
7. Click **Plot**.

The concentration cloud should broaden with time if diffusion is active. The peak should decrease if either diffusion spreads the cloud or annealing removes defects.

31.2 Line plot through the defect cloud

1. Right-click **Results** > **Data Sets**.
2. Choose **Cut Line 2D**.
3. Set the first point to $(0, Ly/2)$.
4. Set the second point to $(Lx, Ly/2)$.
5. Right-click **Results** and choose **1D Plot Group**.
6. Right-click the plot group and choose **Line Graph**.
7. Use the cut-line data set.
8. Set expression to **C**.
9. Plot several times to observe broadening and decay.

31.3 Total defect inventory using an integration operator

Create a domain integration operator.

1. Right-click **Component 1** > **Definitions**.
2. Choose **Component Couplings** > **Integration**.
3. Rename it `intop_dom`.
4. Select **Domain 1**.

Then evaluate the total 2D inventory proxy

$$N_C(t) = \int_{\Omega} C(x, y, t) d\Omega. \quad (31.1)$$

In COMSOL:

1. Right-click **Results** > **Derived Values**.
2. Choose **Global Evaluation**.
3. In **Expression**, type `intop_dom(C)`.
4. Click **Evaluate**.

For zero-flux boundaries and no annealing or source, this quantity should remain nearly constant. If it changes strongly, the boundary condition or coefficient signs are likely wrong.

31.4 Peak concentration

Use **Results** > **Derived Values** > **Maximum** > **Surface Maximum**. Set expression to `C`. Evaluate at all stored times. The maximum should decrease for a freely diffusing Gaussian even without annealing because the same inventory spreads over a broader area.

32 Expanded verification worksheets for Module 3

32.1 Worksheet 1: pure annealing, no diffusion

Set

$$D = 0, \quad S = 0, \quad k = k_{\text{ann}} > 0. \quad (32.1)$$

In the Parameters table set `Ddef=0[m^2/s]`, `S0=0[1/(m^3*s)]`, and `kann=1e3[1/s]`. The exact solution at each point is

$$C(x, y, t) = C(x, y, 0)e^{-kt}. \quad (32.2)$$

Create a line plot of C/C_{init} at the hotspot center and verify exponential decay.

32.2 Worksheet 2: pure diffusion with zero-flux boundaries

Set `kann=0[1/s]` and `S0=0[1/(m^3*s)]`. Keep `Ddef` positive. The concentration should spread out, the peak should decrease, and the total inventory `intop_dom(C)` should remain nearly constant.

32.3 Worksheet 3: uniform concentration preservation

Set `Cinit=Cpeak` by temporarily changing the initial value to a constant. With `kann=0` and `S0=0`, a uniform concentration should remain uniform. Any spatial structure in this test indicates an accidental boundary or source term.

32.4 Worksheet 4: source-only growth

Set $D_{\text{def}}=0[\text{m}^2/\text{s}]$, $k_{\text{ann}}=0[1/\text{s}]$, and $S_0=1\text{e}25[1/(\text{m}^3\cdot\text{s})]$. Start from $C_{\text{init}}=0[1/\text{m}^3]$. The expected solution is

$$C(t) = S_0 t. \quad (32.3)$$

This checks the sign and units of the source term.

33 Expanded imported radiation-map workflow

After the baseline tests, import an initial defect map from Geant4 or MATLAB. The safest file has three columns:

$$x, \quad y, \quad C_{\text{init}}. \quad (33.1)$$

1. Right-click **Global Definitions**.
2. Choose **Functions > Interpolation**.
3. Rename the function **CinitMap**.
4. Select **File** as the data source.
5. Browse to the map file, for example `module3_initial_defect_map.csv`.
6. Set the number of arguments to **2**.
7. Confirm that the columns are interpreted as x , y , and function value.
8. Use **Linear** interpolation for the first test.
9. In **Initial Values 1**, change the initial value from **Cinit** to **CinitMap(x,y)**.
10. Plot **CinitMap(x,y)** before solving.
11. Solve only after the imported map appears at the correct location and magnitude.

Unit warning. If MATLAB exports x and y in micrometers but COMSOL interprets them as meters, the defect cloud will appear in the wrong place or outside the domain. Either export coordinates in meters or set the interpolation argument units carefully.

34 Expanded mesh and time-step convergence tutorial

Diffusion-reaction problems require both spatial and temporal convergence checks.

34.1 Mesh convergence

Run the same pure-diffusion case using three meshes:

Mesh	Maximum element size	Record
Coarse	$\min(\text{sigx}, \text{sigy})/3$	Peak C , inventory, hotspot width.
Medium	$\min(\text{sigx}, \text{sigy})/5$	Same quantities.
Fine	$\min(\text{sigx}, \text{sigy})/8$	Same quantities.

The solution is sufficiently resolved when the medium-to-fine change is acceptably small for the intended project demonstration.

34.2 Time-step check

Use three output spacings:

$$\Delta t_{\text{out}}, \quad \frac{1}{2}\Delta t_{\text{out}}, \quad \frac{1}{4}\Delta t_{\text{out}}. \quad (34.1)$$

Even though COMSOL uses adaptive internal time stepping, the requested output times influence what you inspect and export. If sharp early-time evolution is missed, decrease `dtout`.

35 Expanded export workflow for MATLAB comparison

1. Right-click **Results** > **Export**.
2. Choose **Data**.
3. Select the solution data set.
4. Add expressions `C`, `Duse`, `kuse`, and optionally `intop_dom(C)` through Derived Values.
5. Set the time list to the times you want to compare with MATLAB.
6. Save as `module3_comsol_C_export.csv`.
7. Click **Export**.

In MATLAB, compare the COMSOL concentration field with the MATLAB FEM concentration field at the same physical time and spatial coordinates. For inventory comparison, integrate over the domain in both tools and compare the total value rather than relying only on color plots.

36 Expanded standalone checklist for Module 3

Before trusting the Module 3 COMSOL model, verify the following.

1. The dependent variable is named **C** and has units of concentration.
2. The coefficient **da** is 1.
3. The diffusion coefficient **c** is positive and has units of m^2/s .
4. The reaction coefficient **a** is positive for annealing/removal.
5. The source coefficient **f** has units of $\text{m}^{-3}\text{s}^{-1}$.
6. The initial value is the intended Gaussian, constant, or imported map.
7. The boundaries are zero flux unless a surface sink is intentionally added.
8. The pure annealing case matches exponential decay.
9. The pure diffusion case conserves total inventory under zero-flux boundaries.
10. The uniform concentration case stays uniform.
11. The source-only case grows linearly in time.
12. Mesh refinement does not materially change the reported peak and inventory.
13. Exported COMSOL fields can be compared to MATLAB on a common grid or common set of coordinates.

37 Summary

The Module 3 COMSOL implementation is a click-by-click reference model for the same FEM defect-evolution equation used in MATLAB. The governing equation is

$$\frac{\partial C}{\partial t} = \nabla \cdot (D \nabla C) - k_{\text{ann}} C + S. \quad (37.1)$$

In Coefficient Form PDE, it is entered as

$$\frac{\partial C}{\partial t} + \nabla \cdot (-D \nabla C) + k_{\text{ann}} C = S. \quad (37.2)$$

The essential COMSOL entries are

$$\boxed{d_a = 1, \quad c = D, \quad a = k_{\text{ann}}, \quad f = S.} \quad (37.3)$$

The default verification boundary is zero flux. The most important verification checks are uniform-state preservation, pure diffusion inventory conservation, pure annealing exponential decay, and mesh convergence. Once those pass, imported radiation maps, temperature-dependent coefficients, charged-defect drift, and multi-species reactions can be added in a controlled way.

38 References

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